

CONDITION EVALUATION OF NATURA 2000 TARGET FEATURES

JONÁŠ GAIGR

Nature Conservation Agency of the Czech Republic
Department of Biodiversity Monitoring



WHAT THIS COVERS

- **The question** – is this species, in this site, in the condition the site was designated to keep it in?
- **The data** – what a habitat map and a species record have to carry before either can answer it.
- **The assessment** – what “good” means for a habitat and for a species, and where the threshold it is measured against comes from.
- **The checks** – on the indicator before it is used, and on the result before it is published.
- **The machinery** – the whole evaluation as a public repository anyone can read and re-run.

Worked through on the Czech implementation, so every screenshot and every chart here is a real output rather than a mock-up. That is also why the labels are Czech; each caption carries the translation. The principles are not Czech.





WHY CONDITION IS ASSESSED SITE BY SITE

Article 17 reporting says how a feature is doing in the country. It does not tell the person responsible for one site what to do on Monday.



ADAPTIVE MANAGEMENT, AND WHAT IT ASKS OF DATA

The cycle

1. **Monitor** – collect to a written method, so this year compares with the last.
2. **Evaluate** – set the result against the condition the site is meant to deliver.
3. **Plan and prioritise** – decide where intervention is worth its cost.
4. **Implement** – carry out the management.
5. **Repeat** – and find out whether it worked.

What it needs

- A **target state** that was written down before the result came in.
- An assessment that is **repeatable** – the same data yield the same verdict next year and in the next region.
- A result at the **scale of the decision**: the site, and inside it the subsite that is actually managed.
- **Uncertainty carried through**, not silently resolved – “unknown” is an answer, and it commissions monitoring.



TARGET FEATURES, SITES, SUBSITES

The features

- **61 natural habitat types** of Annex I of the Habitats Directive
- **68 animal** and **41 plant species** of Annex II
- **65 bird species** of Annex I of the Birds Directive

Protection is built out of the combination – habitats, flagship, keystone and umbrella species.

What is evaluated is a target feature **in a site** – not a species in a country, and not a site as a whole.

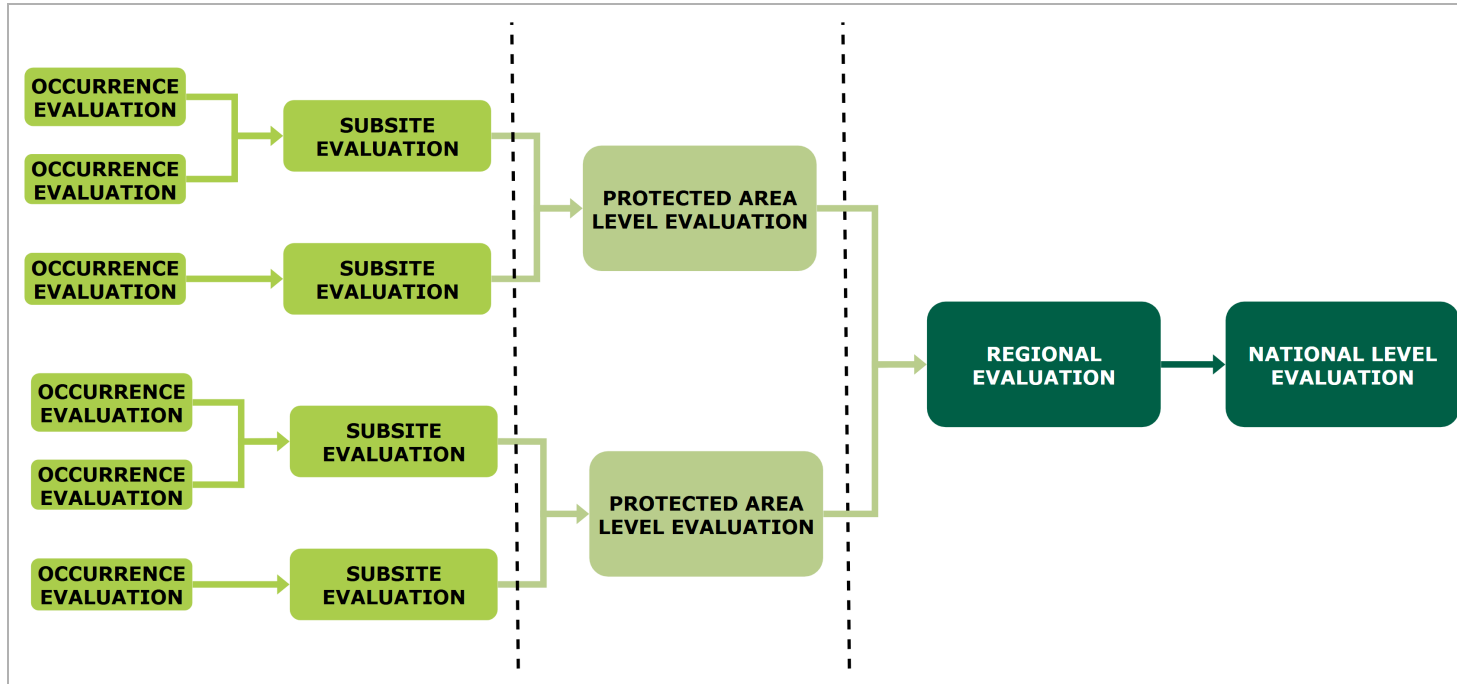
The levels

- A **record** is evaluated where the method allows it.
- Records aggregate to a **subsite** – the pond, the meadow, the cave.
- Subsites aggregate to the **site** (SCI/SAC or SPA): this is the level the result is published at.
- Sites aggregate upward to the **region** and to the **country**.

The chain runs one way. Nothing is assessed at the site that was not assessed below it.



THE CHAIN, DRAWN



Left to right: the assessment of one occurrence; of the subsite it belongs to; of the protected area the result is published at (SCI/SAC or SPA); then the region and the country. Every arrow is a written aggregation rule, not a judgement made afresh at each level. That is what lets a regional total be assembled from sites assessed by different people. Drawn from the framework's own diagram, [BiodivMonCZ/host_naturecz.cz](https://biodivmoncz.host-naturecz.cz/).

FOUR STATES, ONE SCALE

UNKNOWN	GOOD	DETERIORATED	BAD
Insufficient data to decide.	Habitat: sufficient area and quality. Species: stable or increasing population and sufficient habitat.	Habitat: insufficient area or quality. Species: population holding, but habitat insufficient.	Habitat: insufficient area and quality. Species: absent, or population declining.
⇒ monitoring required	⇒ continued occurrence secured	⇒ future occurrence probably threatened	⇒ future occurrence clearly threatened

Four classes, carried unchanged through every level of aggregation and every taxonomic group. A scale that is redefined per group cannot be summed across groups.





THE EVIDENCE BASE

Two national holdings, one rule: if the method is not written down, the comparison is not available.

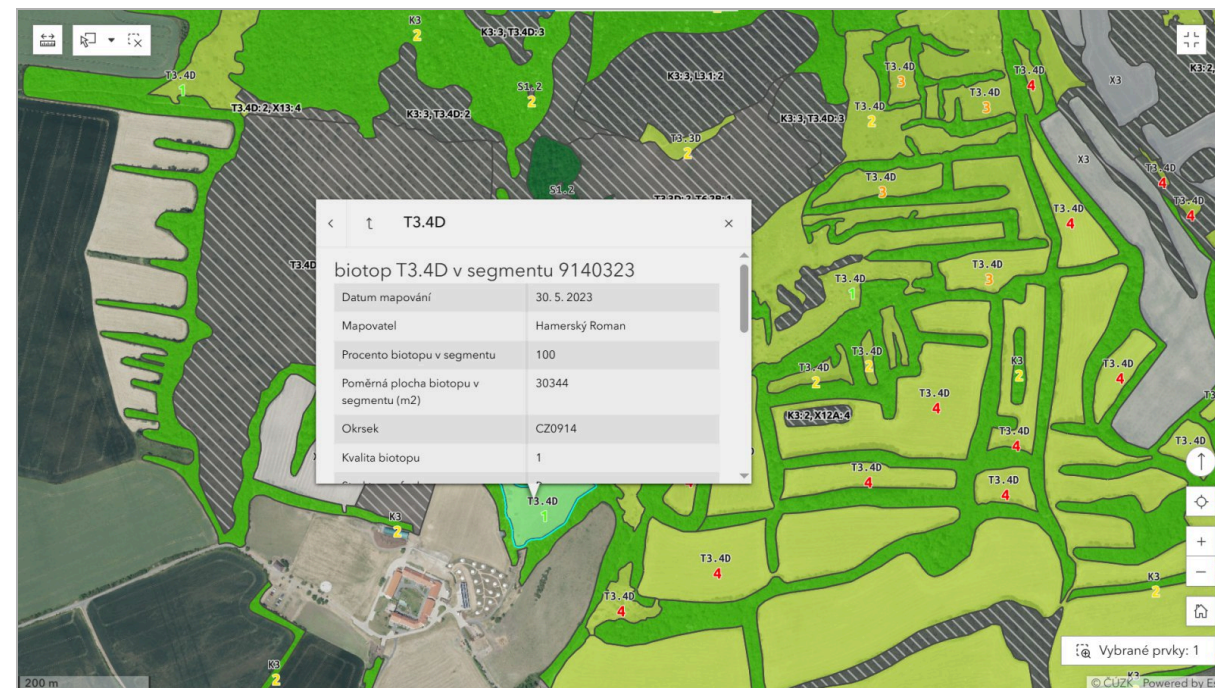


HABITATS: THE MAPPING LAYER

Exact polygons with thin attributes cannot be evaluated at all. The whole country was mapped in **2002–2007** and has been updated since, in roughly **3,500 districts** on a **12-year cycle**.

What each segment has to carry

- **Who** mapped it and **when**. Currency belongs to the segment, not to the layer.
- **How much** of it the habitat occupies, as a share *and* an area, so a mixed segment can still be summed.
- The **quality** score, derived from the field attributes by a published rule.



One segment selected. *Datum mapování* date, *Mapovatel* surveyor, *Procento biotopu v segmentu* share of the segment, *Poměrná plocha* the resulting area in m², *Okrsek* mapping district, *Kvalita biotopu* quality.

TWO PICTURES OF ONE HILLSIDE



The orthophoto. Every boundary a reader draws on it is their own, and no two readers draw the same one.



The mapping layer over the same extent. **T3.4D: 2** – dry grassland at quality 2; hatched grey **X** segments are non-natural. A segment may hold **more than one habitat**, each with its own share and its own quality.

A FIRST HABITAT MAP: THE MINIMUM STANDARD

Map

- **1:10,000** on the orthophoto, in the field. Minimum unit **about 0.25 ha** for extensive types; smaller for rare or high-value stands.
- **Small and linear habitats** – springs, fens, caves – as points or lines, with the **true area or width** recorded.
- **Mosaics** allowed: several habitats per segment, each with its **share** and its own **condition**.
- **Condition as attributes**, scored by a **lookup**, not judged site by site.

Check

- **Re-survey a random 5–10 %** of segments independently; report agreement on the code and on the condition.
- **Points and centroids** say where to map first. They cannot be summed into an area, and Annex III asks for one.
- **Typical species from GBIF**: where to look first, for types with conspicuous indicators. Never an area, a condition or an SDF value.

SPECIES: WHAT A RECORD HAS TO CARRY

- **Which survey it came from.** Only records collected under the relevant methodology set the verdict.
- A **stable identifier**, so a correction can be traced.
- **Absence as data:** **NEG** marks a visit that found nothing – without it a dataset shows arrival and never decline.
- A **date range**, an **observer** and a **survey event**, so effort can be reconstructed.
- A **location at two resolutions** (geometry and grid cell), a **reliability** flag, and **structured observations** rather than prose.

Nález ID	Mapa	Akce	Druh	Lokalizace	Datum od	Datum do	Autor	Kód pole	Věrohodnost	Edit
57380046	3		<i>Triturus cristatus</i> čolek velký NEG	Cihelna Kunín - hliniště, Lenčína tůň	20.7.2023	20.7.2023	Herzogová Klára	6373	1	
57380044			<i>Triturus cristatus</i> čolek velký NEG	Cihelna Kunín - pískovna, úžka	20.7.2023	20.7.2023	Herzogová Klára	6373	1	
57380043	4		<i>Triturus cristatus</i> čolek velký NEG	Cihelna Kunín - hliniště, Štěpánova tůň	16.5.2023	16.5.2023	Herzogová Klára	6373	1	
57380042	3		<i>Triturus cristatus</i> čolek velký NEG	Cihelna Kunín - hliniště, Lenčína tůň	16.5.2023	16.5.2023	Herzogová Klára	6373	1	
57380041	2		<i>Triturus cristatus</i> čolek velký NEG	Cihelna Kunín - hliniště, Nová tůň	16.5.2023	16.5.2023	Herzogová Klára	6373	1	
57380040	2		<i>Triturus cristatus</i> čolek velký NEG	Cihelna Kunín - pískovna, úžka	16.5.2023	16.5.2023	Herzogová Klára	6373	1	
57379976			<i>Triturus cristatus</i> čolek velký	Cihelna Kunín - hliniště, Štěpánova tůň	20.7.2023	20.7.2023	Herzogová Klára	6373	1	
57379975	3		<i>Triturus cristatus</i> čolek velký	Cihelna Kunín - hliniště, Nová tůň	20.7.2023	20.7.2023	Herzogová Klára	6373	1	
54803775			<i>Triturus cristatus</i> čolek velký	Kunín	28.7.2022	28.7.2022	Ožana Stanislav	6373	1	
54803772			<i>Triturus cristatus</i> čolek velký	Kunín	20.6.2022	20.6.2022	Ožana Stanislav	6373	1	
54791327	14		<i>Triturus cristatus</i> čolek velký	Cihelna Kunín - hliniště, Lenčína tůň	12.7.2022	12.7.2022	Lukáš Choleva	6373	1	
54789912	6		<i>Triturus cristatus</i> čolek velký	Cihelna Kunín - hliniště, Nová tůň	14.5.2022	14.5.2022	Lukáš Choleva	6373	1	
54789909	19		<i>Triturus cristatus</i> čolek velký	Cihelna Kunín - hliniště, Nová tůň	11.9.2022	11.9.2022	Lukáš Choleva	6373	1	

Records in the **Species Occurrence Database (NDOP)**. *Nález ID* record id, *Druh* species with the **NEG** badge, *Lokalizace*, *Datum od / do*, *Autor*, *Kód pole* grid cell, *Věrohodnost* reliability. The camera icons count the photographs attached to each record.



HOW A SITE IS ASSESSED

Current state against target state. The difficulty is almost entirely in the second term.



HABITATS: AREA, QUALITY, AND THE TABLE THAT FIXES IT

Area comes from the update mapping, and **degraded segments and clearings count toward it** – raising area and lowering quality rather than erasing the habitat from the ledger.

Quality is derived from the update attributes: **RB** representativeness, **DG** degradation, **SF** structure and function, **RH** regional context.

It is a **lookup**, not a judgement made site by site: the same attributes yield the same value everywhere, every year. The table translates **both vocabularies**, and an unanticipated combination is **added to it**.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	Převodník Pavla Lustyka z 16.12.2019 (email), lehce dopravený Ivou H.														
2															
3	1a) převod parametrů aktualizací a prvního mapování na kvalitu														
4	hodnoty z aktualizací						hodnoty ze základního mapování								
5	kvalita	biotop	RB	DG	SF	RH	REPRE		ZACH	poznámka					
6	1	V-L	F, P, V	0	P		A		A						
7	1	V-L	F, P, V	0	MP		C		A						
8	1	V-L	F, P, V	1	P		B		A						
9	2	V-L	F, P, V	0	N					nově doplněná kombinace					
10	2	V-L	F, P, V	1	MP		D		A						
11	2	V-L	F, P, V	1	N					nově doplněná kombinace					
12	2	V-L	F, P, V	2	P		A		B						
13	2	V-L	F, P, V	2	MP		B		B						
14	2	V-L	F, P, V				C		B						
15	2	V1G	F, P, V			1				nově doplněná kombinace					
16	2	V1G	F, P, V			2				nově doplněná kombinace					
17	2	V4B	F, P, V			1				nově doplněná kombinace					
18	2	V4B	F, P, V			2				nově doplněná kombinace					
19	3	V-L	F, P, V	2	N		A		C						
20	3	V-L	F, P, V	3	MP		D		B						
21	3	V-L	F, P, V				B		C						
22	3	V1G	F, P, V			3				nově doplněná kombinace					
23	3	V4B	F, P, V			3				nově doplněná kombinace					
24	3	vybrané X	F, P, V							nově doplněná kombinace. Vybraná X zatím nejsou definována -> všem X přidělit zatím kvalitu					
25	4	V-L	F, P, V	3	N		C		C						
26	4	V-L	W				D		C						
27	4	V1G	F, P, V			4				nově doplněná kombinace					
28	4	V4B	F, P, V			4				nově doplněná kombinace					
		kvalita VMB	SDF - reprezentativnost	SDF - zachovalost (staré)	SDF - zachovalost										

kvalita 1–4, 1 best, against *hodnoty z aktualizací* the update parameters RB, DG, SF, RH and *hodnoty ze základního mapování* the baseline REPRE and ZACH. *nově doplněná kombinace* – “newly added combination”. The header line says who produced the table and who last adjusted it.

SETTING THE TARGET STATE FOR A HABITAT

The target draws on what was **reported to the Commission** in the Standard Data Form. For **area**, it is one of three:

- the **baseline** mapping value (VMB1),
- the **update** value (VMB2),
- or the **minimum viable area**.

The choice between the first two turns on **why** the figure changed: a methodological change is not a loss.

Quality: the minimum target is **2**. No target may sit below the condition at designation, and every one is reviewed regionally.

Example 1 – methodological change

Baseline 2004: **300 ha** of beech forest. Update: **230 ha**. Minimum viable area: 30 ha.

Regional office judged the drop **methodological**.

→ **Target = 230 ha**

Example 2 – real change

Baseline 2004: **27 ha**. Update: **18 ha**. Minimum viable area: **30 ha**.

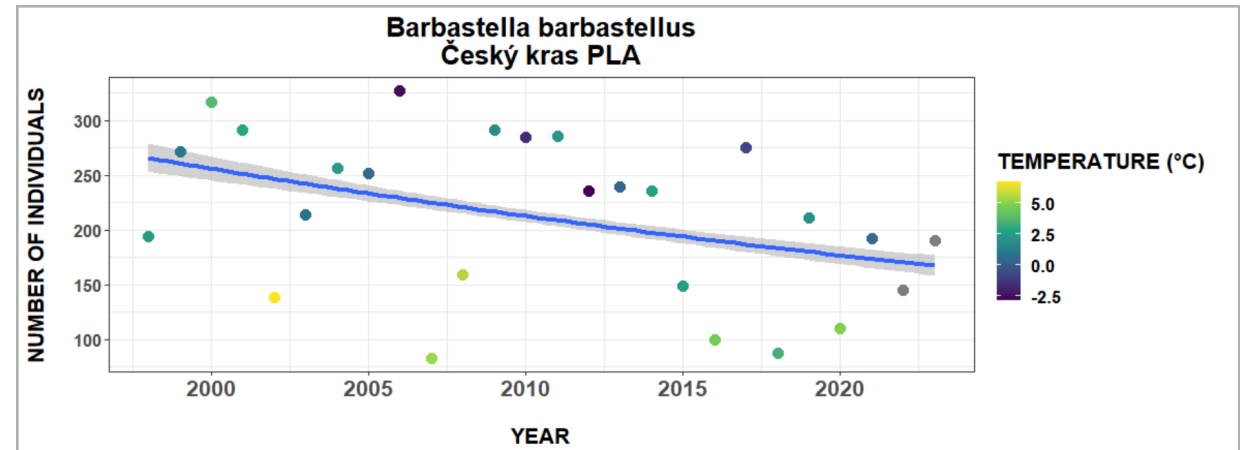
Regional office judged the drop **real**.

→ **Target = 30 ha**

SPECIES: POPULATION FIRST, ENVIRONMENT AS DIAGNOSIS

- The **key indicators are population indicators** – presence, abundance, reproduction, trend.
- **Environmental indicators do not set the verdict.** They say *why* a population indicator came out poorly: habitat loss, fish pressure, water-level manipulation, drying out, encroachment.

The population target is reconciled from **targeted monitoring**, the **Standard Data Form**, the **maximum abundance** in NDOP and **expert knowledge of the site**. Where these disagree, the disagreement is the substance of the discussion.



Population trend for *Barbastella barbastellus* in one site, with mean annual temperature carried on the points. The declining count is the finding; the covariate is where the explanation starts, not where the verdict is made.

FROM MANY INDICATORS TO TWO

The pilot evaluations ran **seven or more indicators at site level**. They were defensible one at a time and nearly unreadable together, and a single weak indicator could carry a whole site.

The revised rule uses **two**:

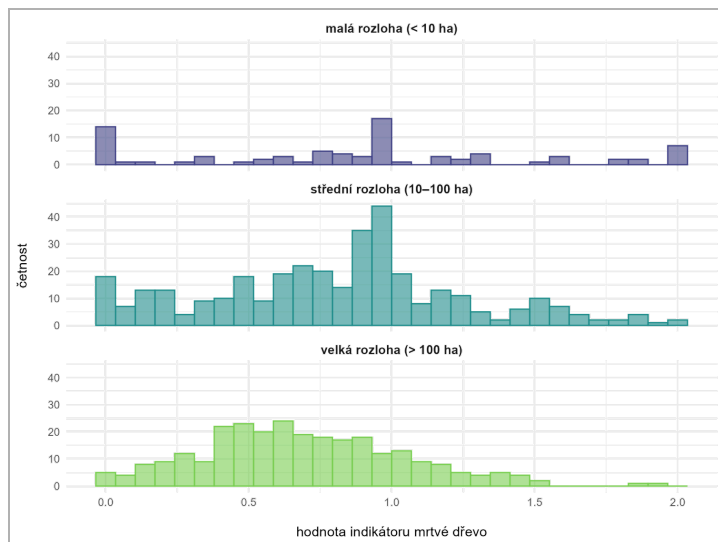
- the **percentage of subsites** assessed as being in good condition,
- the **number of individuals**, as a rolling mean over the last three monitored years, against the target.

Subsites in good condition	Population target	Overall
≥ 70 %	met	GOOD
≥ 70 %	not met	DETERIORATED
< 70 %	met	DETERIORATED
< 70 %	not met	BAD

Piloted on amphibians and extended group by group. A simplified habitat methodology has been through the same reduction.

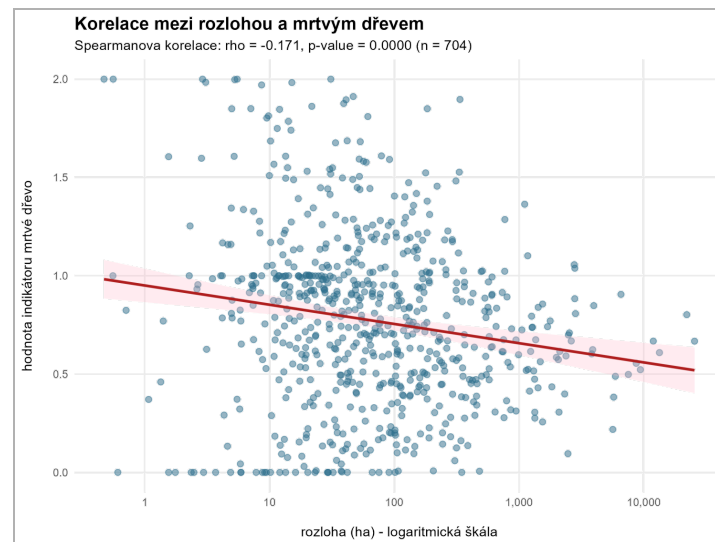


INTERROGATE AN INDICATOR BEFORE YOU THRESHOLD IT



Dead wood, a forest quality indicator, by site area class – *malá* under 10 ha, *střední* 10–100 ha, *velká* over 100 ha. *četnost* is the number of sites.

One threshold for every site would pass and fail partly on size. And **significant is not large**: at $n = 704$, $\rho = -0.17$ clears any p-value you like while accounting for about 3 % of the spread.



The same indicator against site area, log scale. Spearman $\rho = -0.171$, $p < 0.0001$, $n = 704$ – real, and almost nothing.

VALIDATION, AND WHAT PUBLICATION REQUIRES

An **internal process**, in two questions put to the **site manager** or **regional guarantor** – the person who has stood in the site:

1. **Are the input data complete and accurate?** If not – *not validated: faulty data*.
2. **Does the monitoring methodology meet this feature's needs?** If not – *not validated: methodological error*.

Both yes → **validated**. Only validated results are published, at **portal.nature.cz** and carrying their **year**; a class of sites failing the second question is a finding about the **method**.

Název území↑↓	Kategorie CHÚ	🚩	Předmět ochrany↑↓	Kategorie předmětu ochrany	🚩	Rok↓# 1	🚩	Stav	🚩
Třebýčinka u Bezděkova	Evropsky významná lokalita		<i>Bombina variegata</i> kuňka žlutobřichá	Druh		2023		🟢	
Údolí Bánovského potoka	Evropsky významná lokalita		<i>Bombina variegata</i> kuňka žlutobřichá	Druh		2023		🔴	
Přísnotický les	Evropsky významná lokalita		<i>Bombina bombina</i> kuňka obecná	Druh		2023		🟡	
Blovice	Evropsky významná lokalita		<i>Bombina variegata</i> kuňka žlutobřichá	Druh		2023		🟢	
Rašeliníště Radlice	Evropsky významná lokalita		<i>Triturus cristatus</i> čolek velký	Druh		2023		🟢	
Světlik	Evropsky významná lokalita		<i>Leucorrhinia pectoralis</i> vážka jasnoskvrnná	Druh		2023		🟡	
Malý Bukač	Evropsky významná lokalita		<i>Triturus cristatus</i> čolek velký	Druh		2023		🟢	
Pocinovice	Evropsky významná lokalita		<i>Triturus cristatus</i> čolek velký	Druh		2023		🟢	

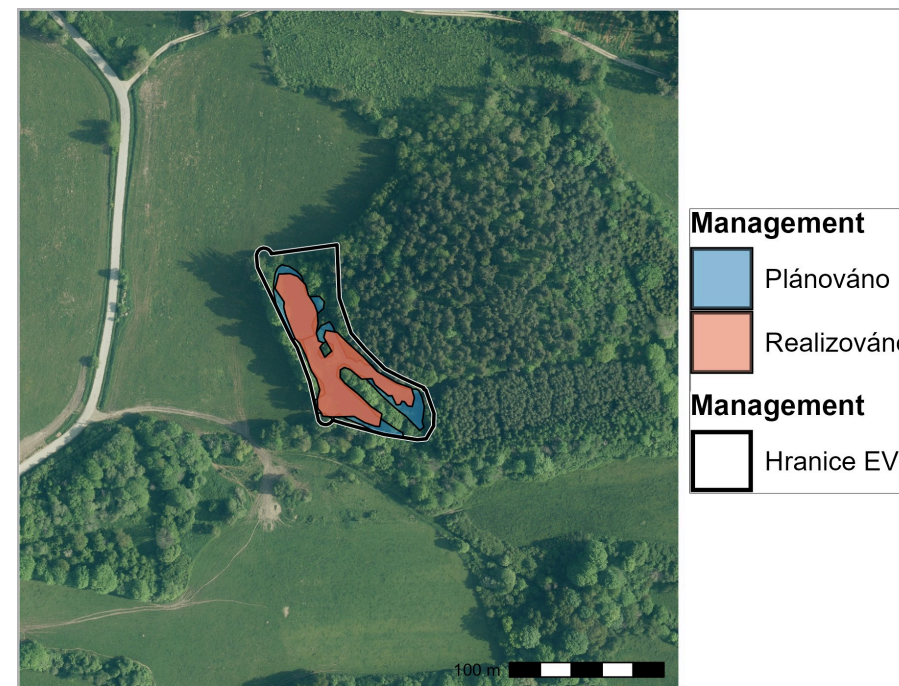
Validated results on the ISOP Portal: site, site category, target feature, feature category, year, condition. Columns are filterable and sortable.

MANAGEMENT EFFECTIVENESS

Condition on its own does not tell you whether management is working. Crossing it with **whether the management plan was actually carried out** does:

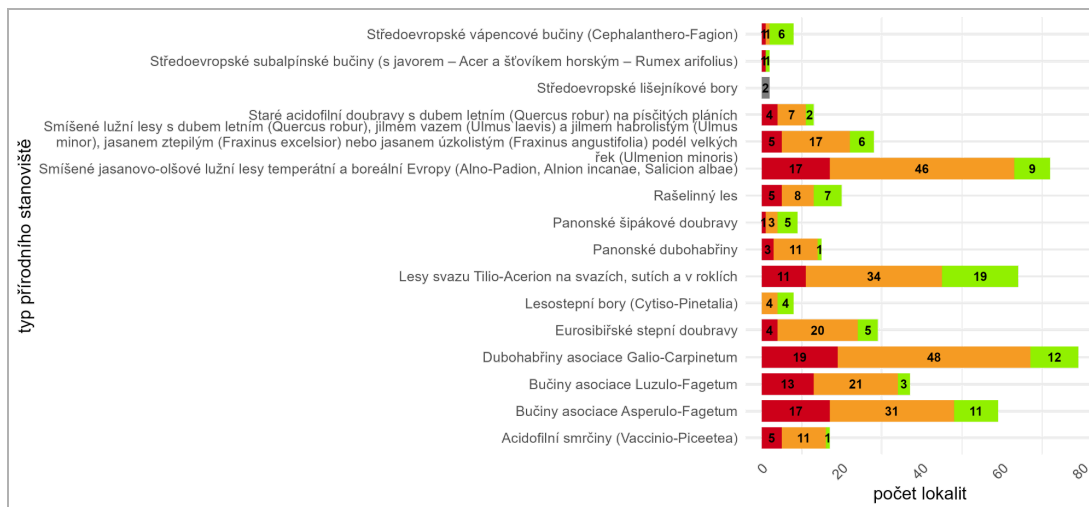
Feature condition	Plan fulfilled	Conclusion
positive	yes	carry on
negative	no	fulfil the plan
positive	no	review the plan
negative	yes	review the plan
insufficient data	–	cannot evaluate

Other influences are carried alongside: nitrogen and sulphur deposition, mean annual temperature, deviation from the climate normal, drought, visitor numbers.

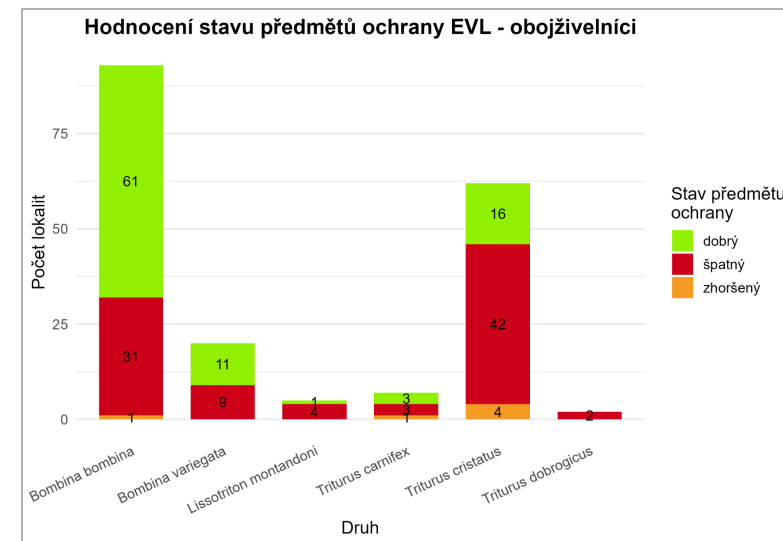


Planned management (*plánováno*) against implemented management (*realizováno*) inside one site boundary (*hranice EVL*). Both are polygons, so the overlap is a number rather than an opinion.

RESULTS



Forest habitat types by number of sites (*počet lokalit*) in each condition class – red bad, orange deteriorated, green good.



Amphibian target features by number of sites in each class: *dobrý* good, *zhoršený* deteriorated, *špatný* bad.

The widespread features carry both the most sites and the most failures, which is where restoration effort has the most to work with, and where the **Nature Restoration Regulation** applies. Amphibians are furthest along – revised methodology, a full pass over the sites, the two-indicator rule applied.

THE EVALUATION RUNS AS CODE

One public R repository, [BiodivMonCZ/host_naturecz](#), applies the written rules to the two national holdings. Nothing is done by hand.

- **Re-runnable.** A correction reaches the published result by re-running, not by editing it.
- **Attributable.** Every rule change has an author, a date and a diff.
- **The cascade is the method.** 21 records → 22 periods → 23 last record → 24 subsites → 25 sites → 27 export; a step that fails stops it.

Thresholds are data, not code. One row per feature × indicator × level:

Column	Carries
DRUH	the target feature
ID_IND	the indicator
TYP_IND	min, max or val
LIM_IND	the threshold
JEDNOTKA	its unit
KLIC	key indicator?
UROVEN	subsite or site

Moving a threshold is an edit to a CSV that can be diffed and reviewed. Nobody edits a function to change a target.

WHAT TRAVELS

- **Write the target down first.** An assessment is only a comparison; without a target set in advance it is a description. And it may not sit below the condition at designation – a target set there locks in the deficit.
- **Put the rule in a table, not in prose.** A lookup can be diffed, reviewed and re-run. A paragraph is re-interpreted by every reader.
- **Compute the spatial relationships; never type them in.** Which site, which zone, which district – all of it follows from one geometry and one query, and survives the next boundary revision.
- **Record absence, effort and observer.** Without them a dataset shows arrival and never decline.
- **Interrogate the indicator before you threshold it** – what it is correlated with, and who scored it. Significant is not large.
- **“Unknown” must be a real outcome.** It is how the system commissions its own monitoring instead of guessing.
- **Validation by someone who knows the site** is what makes the number publishable, and it tests the method at the same time.
- **Keep the evaluation in a repository.** If it is a program rather than a spreadsheet, a correction propagates by re-running and every rule change has an author.

THANK YOU FOR YOUR ATTENTION!
portal.nature.cz



NATURE IS OUR HERITAGE
AND THE FUTURE

aopk.gov.cz

